

Mojtaba Hajian

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Profile

Computational Neuroscientist / Neuroimaging Researcher with a Ph.D. in ICT and an M.Sc. in Biomedical Engineering; specialized in EEG signal processing, ERP analysis, functional connectivity, and machine learning for brain data, with hands-on experience in multimodal neuroimaging (EEG, EEG-fMRI, EEG-fNIRS) and clinical neuroscience applications. Strong background in Python-based neuroimaging pipelines (MNE, EEGLAB, PyRadiomics) and explainable AI for brain and medical data.

Experience

- **SIGNAL PROCESSING AND RECORDING SPECIALIST | NATIONAL BRAIN MAPPING LAB (NBML), IRAN | APRIL 2019 – JANUARY 2022**
 - Led a team as Head of EEG and Signal Processing Department; designed experimental protocols for multi-modal EEG acquisition and analysis
 - Developed protocols for multi-modal EEG data acquisition and signal processing
 - Designed experimental tasks using Psychtoolbox and PsychoPy
 - Analyzed EEG data using EEGLAB, MATLAB Toolboxes, and Python MNE
 - Conducted ERP analysis and EEG-MRI data analysis
 - Managed EEG data recordings for various applications including EEG-VR, EEG-fNIRS, EEG-TMS, and more
- **RESEARCH FELLOW | FREE UNIVERSITY OF BOZEN-BOLZANO, FACULTY OF ENGINEERING, ITALY | JUNE 2025 – PRESENT**
 - Working on the AIRFRAME project on AI-driven radiomics for metastatic castration-resistant prostate cancer (mCRPC), analysing *multi-modal PET/CT images* to predict therapy outcome and survival
 - Build end-to-end pipelines in *Python (PyRadiomics, SimpleITK)* for feature extraction, dimensionality reduction of handcrafted and radiomic features
 - Design and validate *machine-learning and deep-learning models*
 - Implement explainable-AI techniques to visualize feature importances and patient-specific outcome probabilities for clinicians
 - Writing professional reports and publishing in peer-review journals
- **PH.D. CANDIDATE | UNIVERSITY OF CALABRIA, DICES, ITALY | JANUARY 2022 – MARCH 2025**
 - Machine learning & deep learning model development using Keras, Pytorch and Scikit-learn
 - High-dimensional data analysis, statistical learning, and reproducible research pipelines
 - Develop large-scale data analytics workflows (using Apache Spark, Hive, and Hadoop)
 - Leverage unsupervised learning methods (e.g., K-Means, DBSCAN) to discover patterns within complex, high-dimensional data.
 - Author scientific publications, technical reports, and conference papers, effectively communicating research insights to multidisciplinary audiences
 - Utilize Python, Matlab, C#, and Java for data engineering, modeling, and analytical tasks

- **RESEARCH INTERN| ECOLE D'INGÉNIEURS GÉNÉRALISTES, DE L'INFORMATIQUE À L'ÉLECTRONIQUE (ESEO), FRANCE| JUNE 2023 – DECEMBER 2023**
 - Implemented core big data frameworks: Successfully deployed Apache Hadoop, Apache Hive, and Apache Spark to establish a big data processing environment
 - Database Management: Created and managed a TPC-H database for benchmarking OLAP processes, integrating Apache Hive for structured data storage and manipulation
 - OLAP Data Cube Deployment: Deployed and managed an OLAP data cube in Hive, enabling advanced data analysis capabilities
 - Machine Learning Algorithms Deployment: Applied machine learning algorithms over big OLAP data cubes

Skills

- **Neuroscience & Neuroimaging**
 - EEG preprocessing, ERP, time–frequency analysis (EEGLAB, SIFT, FieldTrip, MNE)
 - Functional and effective connectivity (coherence, Granger, wPLI, etc.)
 - EEG-fMRI integration
 - Experimental design (PsychoPy, Psychtoolbox)
- **Computational Methods**
 - Machine learning for neural data (Scikit-learn)
 - Feature engineering & dimensionality reduction
 - Explainable AI (SHAP, feature importance)
- **Deep Learning**
 - CNN, RNN, LSTM, Transformers (TensorFlow, Keras, Pytorch)
 - Deep Graph Neural Networks (GNNs), Reinforcement Learning (RL)
- **Programming**
 - Python, Matlab, Java, C#
- **Big Data & Cloud Computing**
 - Apache Spark, Hive, Hadoop
 - AWS, GCP

Education

- **PH.D. IN INFORMATION AND COMMUNICATION TECHNOLOGY (ICT) | MARCH 2025| UNIVERSITY OF CALABRIA, ITALY**
Dissertation: *Multidimensional Big Data Analytics for Knowledge Discovering and Analysis from Big Web Knowledge Bases*
Supervisor: Prof. Alfredo Cuzzocrea
- **M.S. IN BIOMEDICAL ENGINEERING | JULY 2017 | AMIRKABIR UNIVERSITY OF TECHNOLOGY, IRAN**
Dissertation: *Quantification of Depression Disorder Using EEG Signal*
 - EEG preprocessing, feature extraction, ML-based biomarkers**Supervisor:** Prof. Mohammad Hassan Moradi
- **B.S. IN ELECTRONICS ENGINEERING | SEPTEMBER 2014 | BABOL NOSHIRVANI UNIVERSITY OF TECHNOLOGY, IRAN**
Dissertation: *Design and Implementation of A Digital Compass*
Supervisor: Prof. Mehdi Ezoji

Publications

Submitted or In Progress

- [In Progress] Mojtaba Hajian, Somayeh Khosravi Khorashad, Sepideh Khoneiveh, Reza Khosrowabadi, **“Decoding Neural Dynamics of Suspense: EEG Connectivity and Power Analysis with Machine Learning for Pre- and Post-Suspense State Classification”**, *Applied Soft Computing*, 2026.
- Alfredo Cuzzocrea, Mojtaba Hajian, **“Multidimensional Clustering over Big Data: Models, Issues, Analysis, Emerging Trends”**, *Journal of Big Data*, Springer, 2026.
- [In Progress] Alfredo Cuzzocrea, Mojtaba Hajian, **“Data Cube Compression at Work! Effective and Efficient OLAP over Big Datasets in Mobile Cloud Environments”**, *Information Systems*, 2026.

Conference & Journal Papers

- Mojtaba Hajian, Mohammad Hassan Moradi, **‘Quantification of Depression Disorder Using EEG Signal’**, *The 24th National and the 2nd International Iranian Conference on Biomedical Engineering (ICBME 2017)*, Tehran, Iran, 2017.
- Yousef Mohammadi, Mojtaba Hajian, Mohammad Hassan Moradi, **‘Discrimination of Depression Levels Using Machine Learning Methods on EEG Signals’**, *The 27th Iranian Conference on Electrical Engineering (ICEE 2019)*, Yazd, Iran, 2019.
- Somayeh Khosravi Khorashad, Sepideh Khoneiveh, Mojtaba Hajian, Reza Khosrowabadi, **‘Decoupage Parameters Related to the Suspense Points and Symmetrical Activation of Brain Waves at the Frontal Lobe: An EEG Study of the Movie ‘Psycho’ ’**, *Quarterly Review of Film and Video*, 2022.
- Alfredo Cuzzocrea, Mojtaba Hajian, **“ClassCube: Effective and Efficient Big OLAP Data Cube Classification via Dimensionality Reduction”**, *ICEIS*, 2024.
- Alfredo Cuzzocrea, Mojtaba Hajian, Abderraouf Hafsaoui, **“MALAGA – MultidimensionAL Big Data Analytics over Massive Graph Data”**, *IEEE International Conference on Big Data (2024)*, Washington DC, USA, 2024.
- Alfredo Cuzzocrea, Mojtaba Hajian, **“Experimental Analysis and Assessment of a Real-Life Cloud Mobile Big OLAP System Enhanced with Compression and Approximation Paradigms”**, *IEEE International Conference on Big Data (2024)*, Washington DC, USA, 2024.
- Alfredo Cuzzocrea, Mojtaba Hajian, Abderraouf Hafsaoui, **“Compressing Big OLAP Data Cubes over Mobile Clouds: A Hierarchy-Based Data Partitioning Approach”**, *Proceedings of the 32nd Symposium of Advanced Database Systems, Villasimius, Italy*, 2024.
- A Cuzzocrea, M Hajian, **‘Effective and Efficient Big OLAP Data Cube Compression in Mobile Cloud Environments: The IQTS Algorithm’**, *IEEE International Conference on Big Data (2023)*, Sorrento, Italy.
- A Cuzzocrea, P Figuera, M Hajian, PG Bringas, **‘A Theoretical Framework for Supporting Clustering Validation via Non-Negative-Matrix-Factorization Trace Sequences Over Probabilistic Spaces’**, *IEEE Conference on Cloud and Big Data Computing (CBDDCom 2023)*, Abu Dhabi, United Arab Emirates.
- Alfredo Cuzzocrea, Carson Leung, Mojtaba Hajian and Marshall Jackson, **‘Effectively and Efficiently Supporting Predictive Big Data Analytics over Open Big Data in the Transportation Sector: A Bayesian Network Framework’**, *The 8th IEEE International Conference on Cloud and Big Data Computing (CBDDCom 2022)*, Calabria, Italy.

- Alfredo Cuzzocrea, Christophe Cerisara, Mojtaba Hajian, ‘**Risk Analysis for Unsupervised Privacy-Preserving Tools**’, *Proceedings of the 30th Italian Symposium on Advanced Database Systems, SEBD 2022, Tirrenia (Pisa), Italy, June 19-22, 2022*.

Teaching Experiences

- “Machine Learning Applications for Biomedical Signal Processing”, Webinar, Brain Mapping Laboratory, Tehran, Iran, February 2021.
- “The 16th workshop on EEG Signal Recording, Processing and Analyzing”, Workshop, National Brain Mapping Laboratory, Tehran, Iran, October 2020.
- “Instructor of over 50 trainees in the EEG data analysis and recording internship course”, Internship, National Brain Mapping Laboratory, Tehran, Iran, August 2019-December 2021.
- “An introduction to the function of the EEG amplifier and how to use it”, Seminar, Brain Mapping Laboratory, Tehran, Iran, January 2020.
- “Robotics and micro-controllers” Workshop, Babol Noshirvani University of Technology, Mazandaran, Iran, July 2010-September 2013.

Honors and Awards

- Commendation for Establishing EEG Setup for Sleep Services
 - National Brain Mapping Laboratory (NBML) | 2020
- Team Leader Award for Manufacturing Electronic MRI-Compatible Response Grips
 - National Brain Mapping Laboratory | 2020
- Reviewer for 'Frontiers in Biomedical Technologies' Journal and 'IEEE Big Data' Conference
- 2nd Place in Smart City Startup Weekend (MOPEC-2017)
 - Sponsored by Municipality of 5th District of Tehran City | 2017

References

- Dr. Floriano Zini
 - Senior Researcher of Computer Science
 - Free University of Bozen-Bolzano
 - Email: floriano.zini@unibz.it
- Prof. Alfredo Cuzzocrea
 - Professor of Computer Engineering
 - University of Calabria
 - Email: alfredo.cuzzocrea@unical.it
- Prof. Slimane Hammoudi
 - Professor at Computer Science Department
 - ESEO, France
 - Email: Slimane.HAMMOUDI@eseo.fr
- Prof. Reza Khosrowabadi
 - Professor at the Institute for Cognitive and Brain Sciences
 - Shahid Beheshti University
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